CHAPTER 10

Active Support's Concern Module

In the previous chapter, you saw that the modules in Rails are special: when you include them, you gain both instance and class methods. How does that happen?

The answer comes from yet another module: Concern, in the Active Support library. ActiveSupport::Concern twists and bends the Ruby object model. It encapsulates the "add class methods to your includer" functionality, and it makes it easy to roll that functionality into other modules.

ActiveSupport::Concern is easier to understand if you know how it came to exist in the first place. We'll start by looking back at Rails' older versions, before Concern entered the scene.

Rails Before Concern

The Rails source code has changed a lot through the years, but some basic ideas haven't changed much. One of these is the concept behind ActiveRecord::Base. As you've seen in *ActiveRecord::Base*, this class is an assembly of dozens of modules that define both instance methods and class methods. For example, Base includes ActiveRecord::Validations, and in the process it gets instance and class methods.

The mechanism that rolls those methods into Base, however, has changed. Let's see how it worked in the beginning.

The Include-and-Extend Trick

Around the times of Rails 2, all validation methods were defined in ActiveRecord::Validations. (Back then, there was no Active Model library.) However, Validations pulled a peculiar trick: